

Study Guide for Motion Test

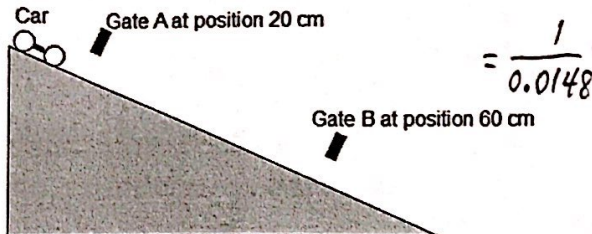
Name: **KEY**

Terms (Able to Define the following. Include units, and if it is a vector or not) Reference point/origin, Position, Displacement, Distance, Speed, Vector vs. Scalar, Average Velocity, Instantaneous Velocity, Acceleration.

Car & Ramp Labs

Width of wing on car = 1 cm
(popsicle stick)

Time in Gate A = 0.0148 s
Time in Gate B = 0.0058 s
Time from A-B = 0.3496 s



- Find the Displacement AB $60 - 20 = 40 \text{ cm}$
(Show the numbers you use before you solve)
- Find the Average Velocity A to B (V_{AB}) = $\frac{d}{t} = \frac{40}{0.3496} = 114 \text{ cm/s}$
- Find the Instantaneous Velocity (V_A) = 67.6
- Find the Instantaneous Velocity (V_B) = $\frac{1}{0.0058} = 172 \text{ cm/s}$
- Find the acceleration down the ramp = $\frac{172 - 67.6}{0.3496} = 299 \text{ cm/s/s}$

Lab Answer Check 40 67.6 114 172 299
(cm, cm/s, cm/s/s)

Mixed Problem Solving Practice (List knowns and unknowns, Show Equation, Substitute & Solve!) See Equations in notes....

- A dancer moves from position zero to position 8 m and then back to position 3. The entire motion requires 10 seconds to complete. (sketch may help here)
 - Find the distance travelled $8 + 5 = 13 \text{ m}$
 - Find the displacement 3 m
 - Find the average velocity (w/ magnitude, units and direction) $v = \frac{d}{t} = \frac{3}{10} = 0.3 \text{ m/s fwd}$
- The flight speed of a European swallow is 11 m/s. How long will it take for the bird to travel 1600 meters? $\frac{1600}{11}$
- A car is driving on the frustrating 55 freeway. It moves 500 meters in 30 seconds, 200 meters in 60 s, and 100 meters in 120 s. What is the average velocity of the car? $\frac{d}{t} = \frac{500 + 200 + 100}{30 + 60 + 120} = \frac{800 \text{ m}}{210 \text{ s}} = 3.81 \text{ m/s}$
- A ball rolls down a hill at a rate of 2.0 m/s/s. The ball starts from rest and travels for 4 s.
 - How far did the ball travel? $a = 2 \quad v_i = 0 \quad t = 4 \quad d = v_i t + \frac{1}{2} a t^2 = 0 + \frac{1}{2} (2) (4)^2 = 16 \text{ m}$
 - What was the ball's velocity at $t = 4 \text{ sec}$? $v_f = v_i + a t = 0 + (2) (4) = 8 \text{ m/s}$
- A car initially moving at 25 m/s hits the brakes and slows at a rate of 3 m/s/s (pos or neg?). If he brakes for 5 seconds, find the displacement of the car. $v_i = 25 \quad a = -3 \quad t = 5 \quad d = ? = v_i t + \frac{1}{2} a t^2$
- A rock climber's shoe loosens a rock, and her climbing buddy at the bottom of the cliff notices that the rock takes 3.2 s to fall to the ground. $v_i = 0 \quad a = 9.8 \quad t = 3.2$
 - How high up the cliff is the rock climber? $d = v_i t + \frac{1}{2} a t^2 \rightarrow d = 0 + \frac{1}{2} (9.8) (3.2)^2$
 $d = 50.2 \text{ m}$
 - How fast was the rock moving just before impact?
- Mike Trout hits a foul ball straight up. If the ball leaves the bat at 30 m/s, find the time the ball is in the air (if it comes back down to the same place).

Answer Bank 0.3 1.2 3 3.81 6.1 8 13 16 31.4 35 50.2 87.5 145

$$v_f = v_i + a t$$

$$-30 = 30 + (-9.8) t$$

$$-60 = -9.8 t$$

$$6.1 \text{ sec} = t$$

$$v_f = v_i + a t$$

$$v_f = 0 + (9.8) (3.2)$$

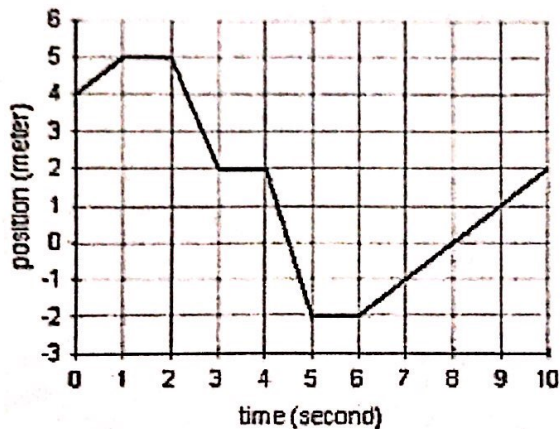
$$v_f = 31.4 \text{ m}$$

$$d = v_i t + \frac{1}{2} a t^2$$

$$d = 25(5) + \frac{1}{2} (-3)(5)^2$$

$$d = 125 - 37.5$$

$$d = 87.5 \text{ m}$$



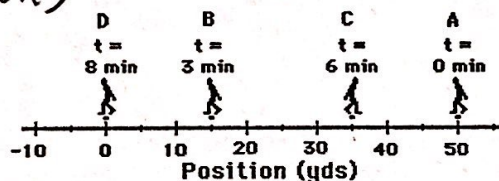
Position vs. Time Graph

1. What is the initial position? 4 m
2. What is the distance travelled from 0 to 10 seconds? $1 + 7 + 4 = 12\text{ m}$
3. What is the displacement from 0-10 sec? -2 m (bwd)
4. Which time interval has greatest speed? $4-5$
5. Which time interval(s) are not moving? $1-2, 3-4, 5-6$
6. When does object change direction? $1-2, 5-6$
7. Find velocity during time interval 4-5 sec. $\rightarrow \frac{-4\text{ m}}{1\text{ s}} = -4$
8. Find velocity during time interval 6-10 sec. $\rightarrow \frac{2 - (-2)}{4} = \frac{4}{4} = 1\text{ m/s}$

$$v = \frac{d}{t} = \frac{2 - (-2)}{4} = \frac{4}{4} = 1\text{ m/s}$$

Which of the following statements about distance and/or displacement are TRUE? List all that apply.

1. A person makes a round-trip journey, finishing where she started. The displacement for the trip is 0 and the distance is some nonzero value. **T**
2. The phrase "20 m, west" likely describes the displacement for a motion. **T**
3. The diagram below depicts the path of a person walking to and fro from position A to B to C to D. The distance for this motion is 100 yds. $35 + 20 + 35 = 90\text{ yds}$
4. For the same diagram below, the displacement is 50 yds. -50 yds (direction)



Which of the following statements about velocity, speed & acceleration are TRUE? List all that apply.

1. Velocity is a vector quantity and speed is a scalar quantity. **T**
2. Both speed and velocity refer to how fast an object is moving. **T**
3. Person X moves from location A to location B in 5 seconds. Person Y moves between the same two locations in 10 seconds. Person Y is moving with **half the speed** as person X. **T**
4. The velocity of an object refers to the rate at which the object's position changes. **T**
5. The phrase "30 mi/hr, west" likely refers to a scalar quantity. **F**
6. An object which is slowing down has an acceleration. **T**
7. An object which is moving at constant speed in a circle has an acceleration. **T**
8. Acceleration is the rate at which the velocity changes. **T**
9. An object that is accelerating is moving fast. **F** (not necessarily)

On the velocity-time graph below, sketch a plot representing the motion of an object which is Label each line with the corresponding letter (e.g., "a", "b", "c", etc.) Positive is forward, negative is backward

- a. at rest
- b. moving in the positive direction at constant speed
- c. moving in the negative direction from slow to fast
- d. moving in the negative direction from fast to slow
- e. moving with a positive velocity and a positive acceleration
- f. moving with a positive velocity and a negative acceleration
- g. moving with a positive velocity at constant speed and then decelerating to a stop

